



सत्यमेव जयते

NITI Aayog

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SECTOR AT A GLANCE

Medium to Long-Term Training

Skilling, Livelihoods and Labour Welfare Sector in India

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Labour Welfare Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

The medium to long-term Vocational Education and Training (VET) is the foundation of skill development ecosystem in India. The first Polytechnic and Industrial Training Institute (ITI) was established in the first half of the 20th century. Both ITIs and Polytechnic institutes (Government and private) are primarily governed through Directorate General of Training (DGT) and together provide VET to millions of youth in the country across sectors and industry occupations. Broadly medium to long term training is classified under 3 categories:

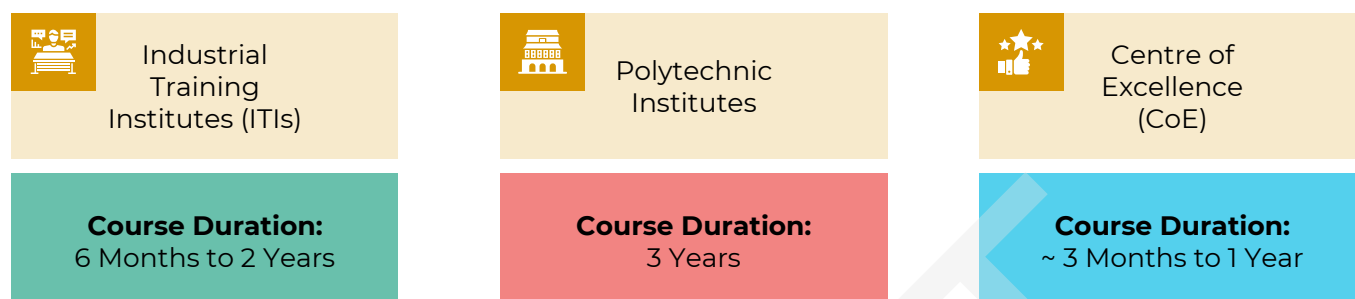


Figure 1: Overview of medium to long term training ecosystem in India

Industrial Training Institutes (ITIs)

First ITI was established in 1950 under Craftsmen Training Scheme (CTS) with an objective to meet the workforce needs of various industries and to promote industrialisation in India. ITIs are governed by DGT w.r.t norms and standards while the day-to-day management rests with concerned State directorates/departments. Trainees opt for ITI courses after passing 10th/12th.

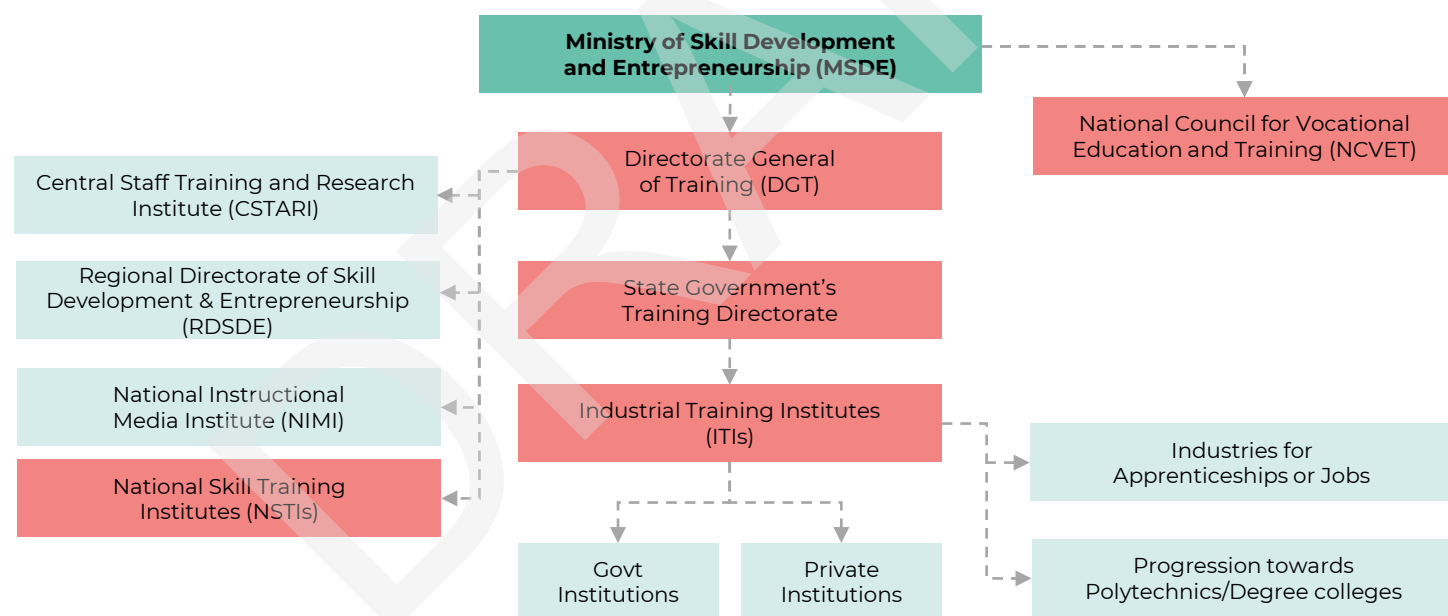
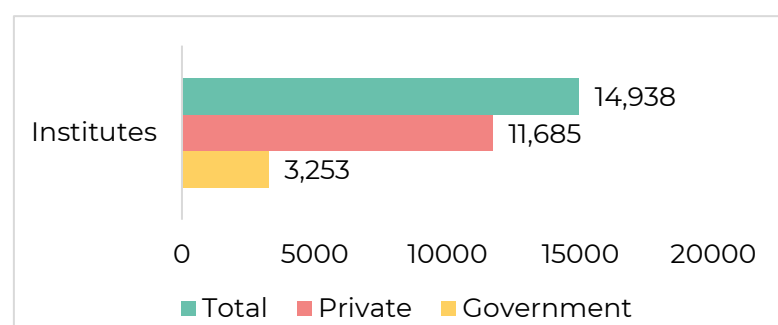


Figure 2: ITI ecosystem in India



Out of total **14,938 ITIs** in India, **21.78% (3,253)** are Government funded while **78.22% (11,685)** are private institutions.¹

Uttar Pradesh has maximum ITIs at **21.61% (3,229)**²

Figure 3: ITI institutes spread in India across Government and private sector

¹ Directorate General of Training (DGT) - NCVT MIS Portal. Retrieved 12 July 2023

² Directorate General of Training (DGT) - NCVT MIS Portal. Retrieved 12 July 2023

Polytechnic Institutes

Polytechnics in India impart long-term diploma courses in various branches (engineering and non-engineering) which can be opted by 10th/12th pass-out students. The Polytechnic institutes (both Government and private) are regulated by All India Council of Technical Education (AICTE) and managed by DGT and respective State directorates/departments.

As of now, for academic year 2022 – 2023, 3598 institutes have been affiliated by AICTE for offering diploma programs with an annual intake capacity of approx. 9.67 lakh candidates.³

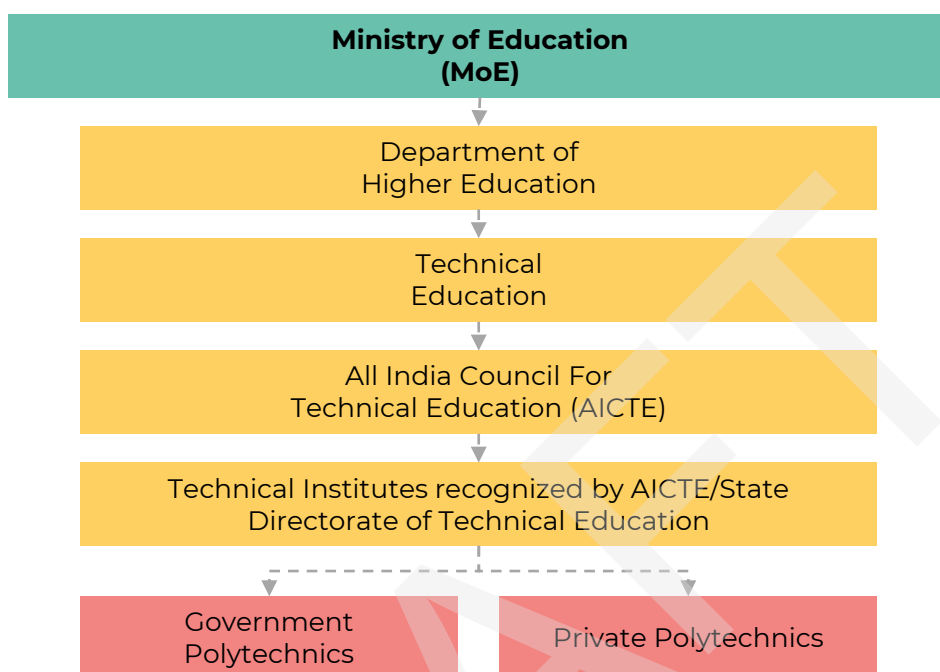


Figure 4: Polytechnic ecosystem in India

Centre of Excellence

Centre of Excellence (CoE) in the skilling ecosystem is envisioned to be a model center which is either established or actively operated in partnership with industry to standardise the quality of training, improve outcomes, cater to the industry demand of workforce and emerge as a benchmark for futuristic skill training. Ministry of Skill Development and Entrepreneurship (MSDE) has undertaken an initiative to recognize eligible skill institutes as CoEs with an objective to provide a continuous supply of skilled workforce and propagate best practices across skilling value chain.

Existing Models of Centre of Excellence envisaged by MSDE



Figure 5: Existing Models of Centre of Excellence envisaged by MSDE

Apart from MSDE, other ministries like Ministry of Micro, Small & Medium Enterprises (MSME) and Public Sector Units (PSUs) under Ministry of Petroleum & Natural Gases (MoP & NG) have come forward to establish advanced tool rooms/ technology centers serving specific industry sectors/clusters across country.

³ All India Council of Technical Education (AICTE). Retrieved 18 July 2023

Snapshot of recognised Centers of Excellence (CoEs)

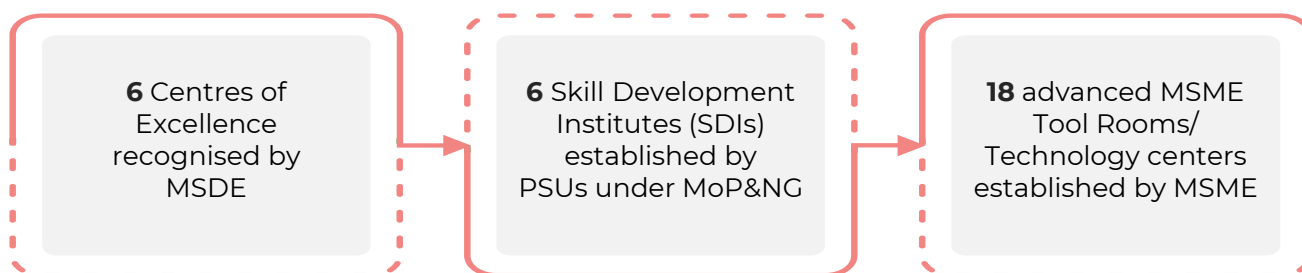


Figure 6: Recognised Centres of Excellence

Sub-Sectoral Trends

Industrial Training Institutes - State wise distribution

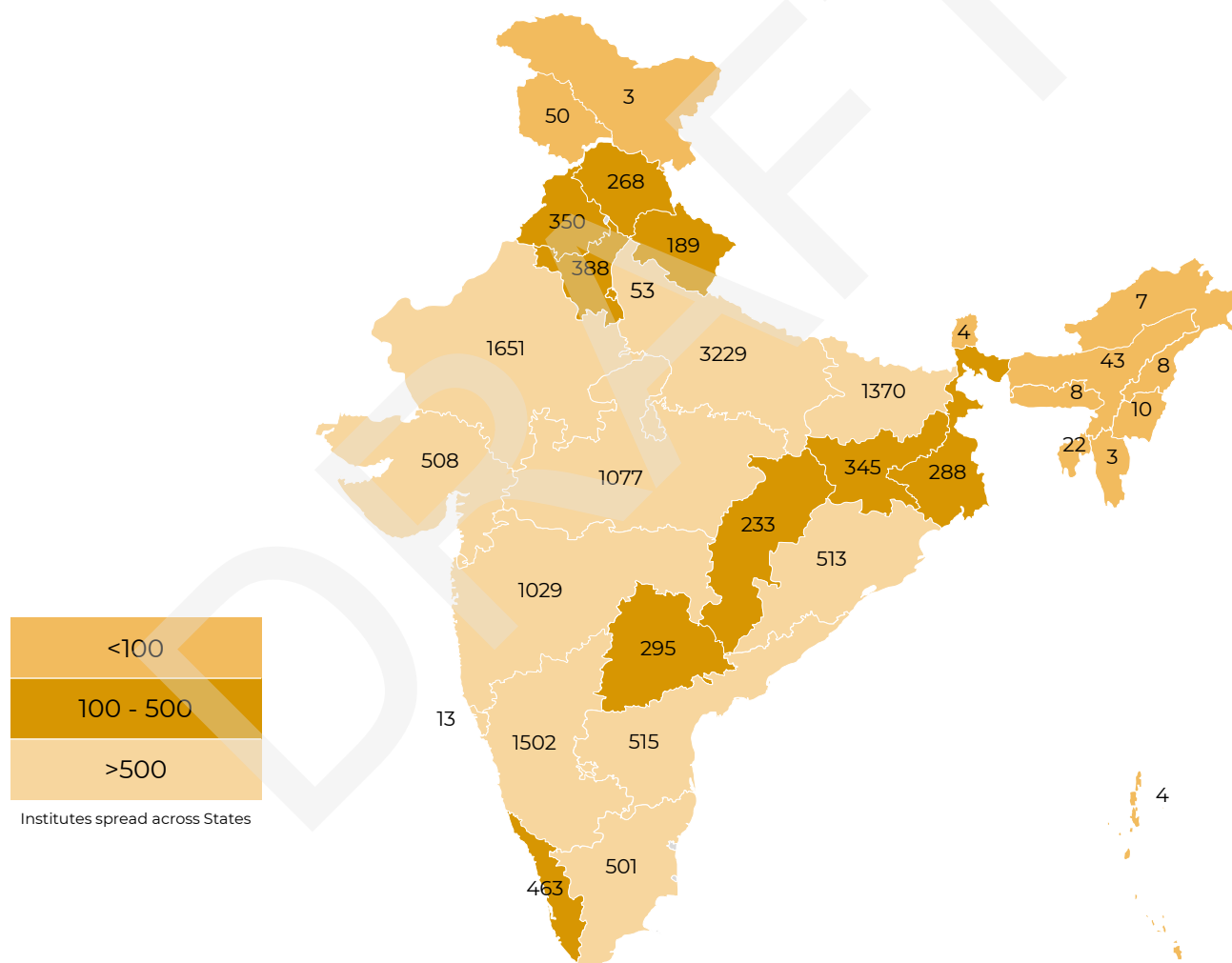


Figure 7: ITI institutes spread in India across different States⁴

- Around **66%** of the total **14,938** ITIs, are situated in **5 States** i.e., Uttar Pradesh, Rajasthan, Karnataka, Bihar, Madhya Pradesh, and Maharashtra.
- Uttar Pradesh has the highest number of total ITIs at **21.61% (3,229)**.
- Maharashtra has the highest number of Government ITIs (**12.98%**).

⁴ Directorate General of Training (DGT) - NCVT MIS Portal - ITI. Retrieved 12 July 2023

Growth of ITIs in the country

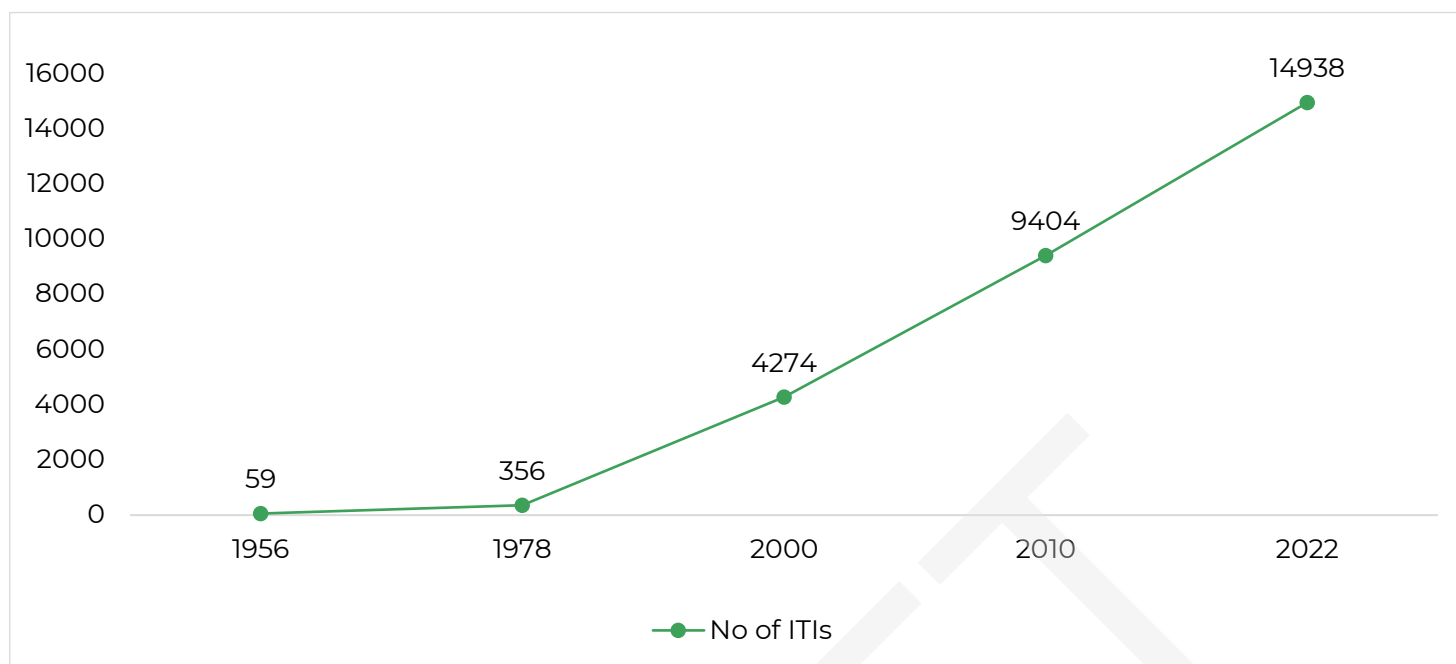


Figure 8: Growth of ITI institutes in India

- Since inception, number of ITIs have increased by more than 200 times in India.
- In last decade, there is an overall growth of **~58%** in number of ITIs established in India. ⁵

Top 10 trades by admitted students

In academic session 2022, out of admitted 12,42,109 students in ITIs, approx. 433K (35%) have enrolled for electrician course followed by fitter (205K - 16.52%) ⁶

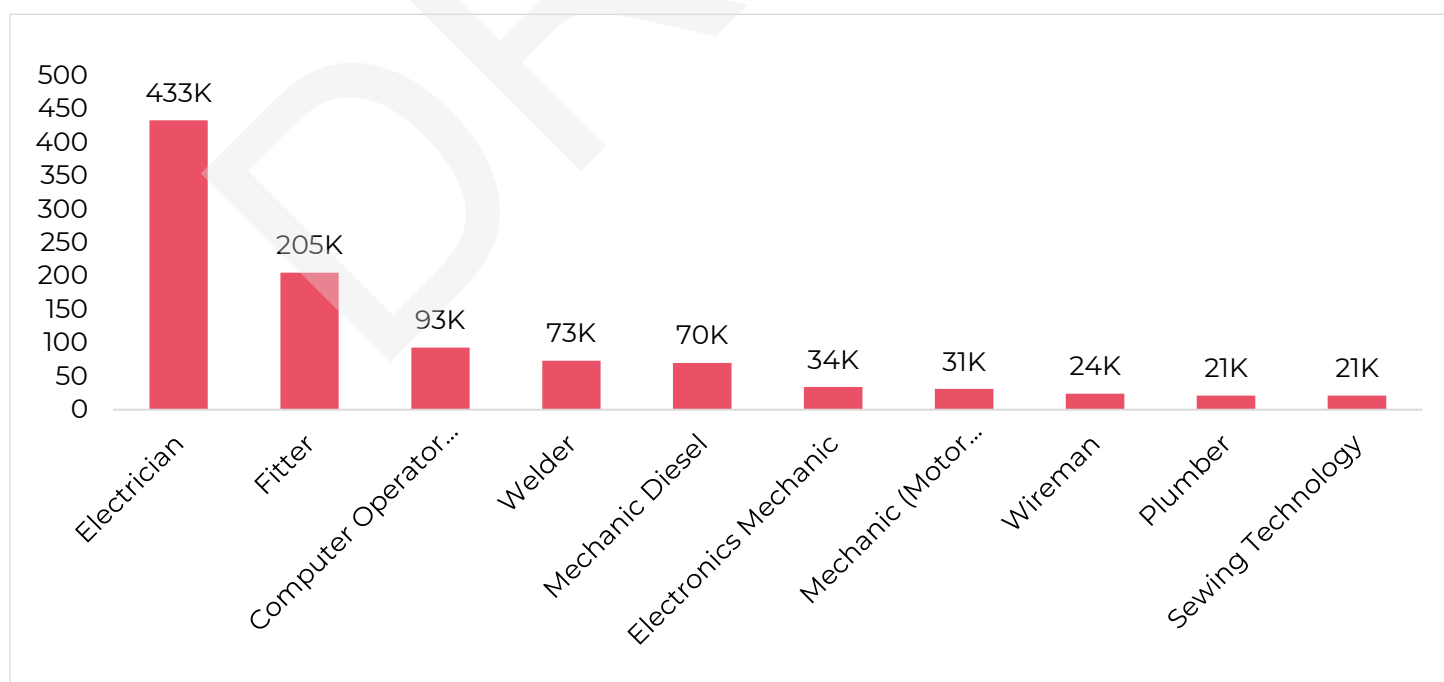


Figure 9: Top 10 trades adopted by admitted students in ITI institutes across India

⁵ Directorate General of Training (DGT) - NCVT MIS Portal - ITI. Retrieved 12 July 2023

⁶ Directorate General of Training (DGT) - NCVT MIS Portal - Admitted Trainee. Retrieved 12 July 2023

Enrolment Distribution - Gender

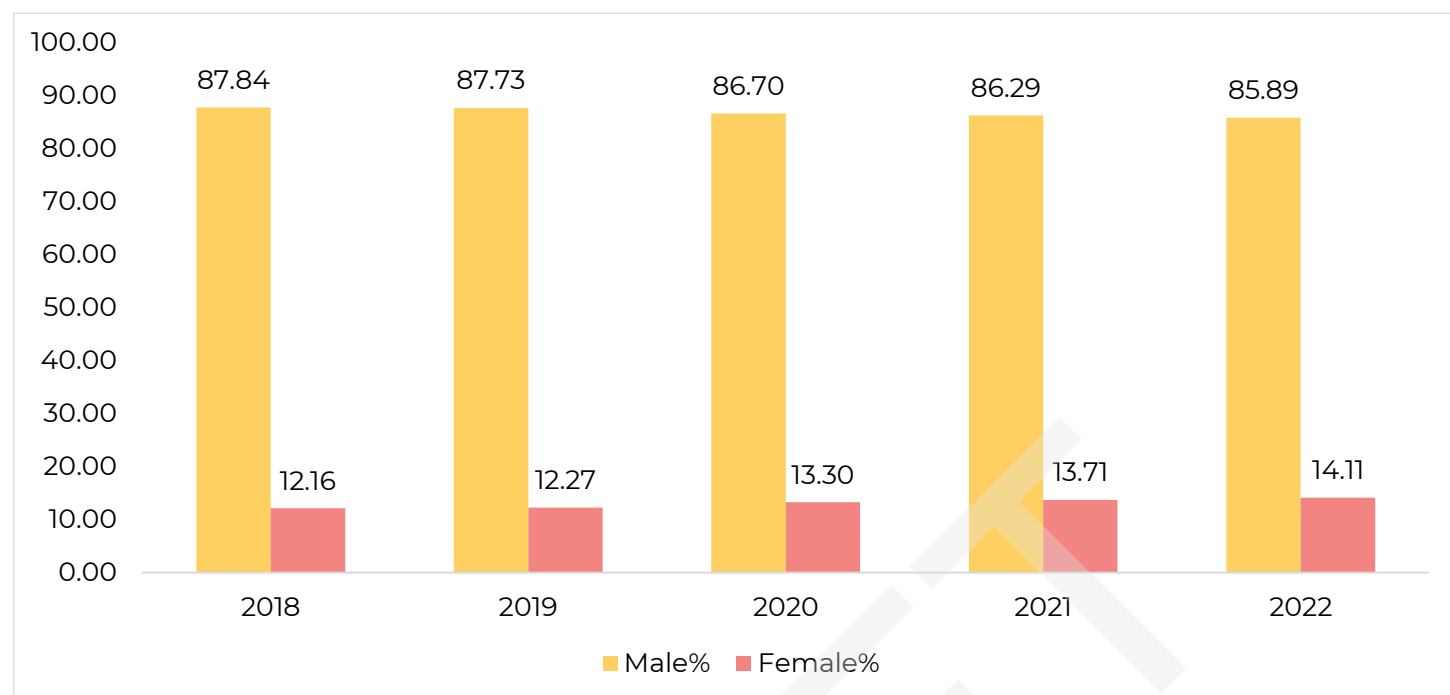


Figure 10: Enrolment distribution of Male and Female across ITI institutes in India

Female representation in ITI enrolments has increased marginally over a period of 2018 – 2022 ⁷

Seat Utilisation Across Institutes

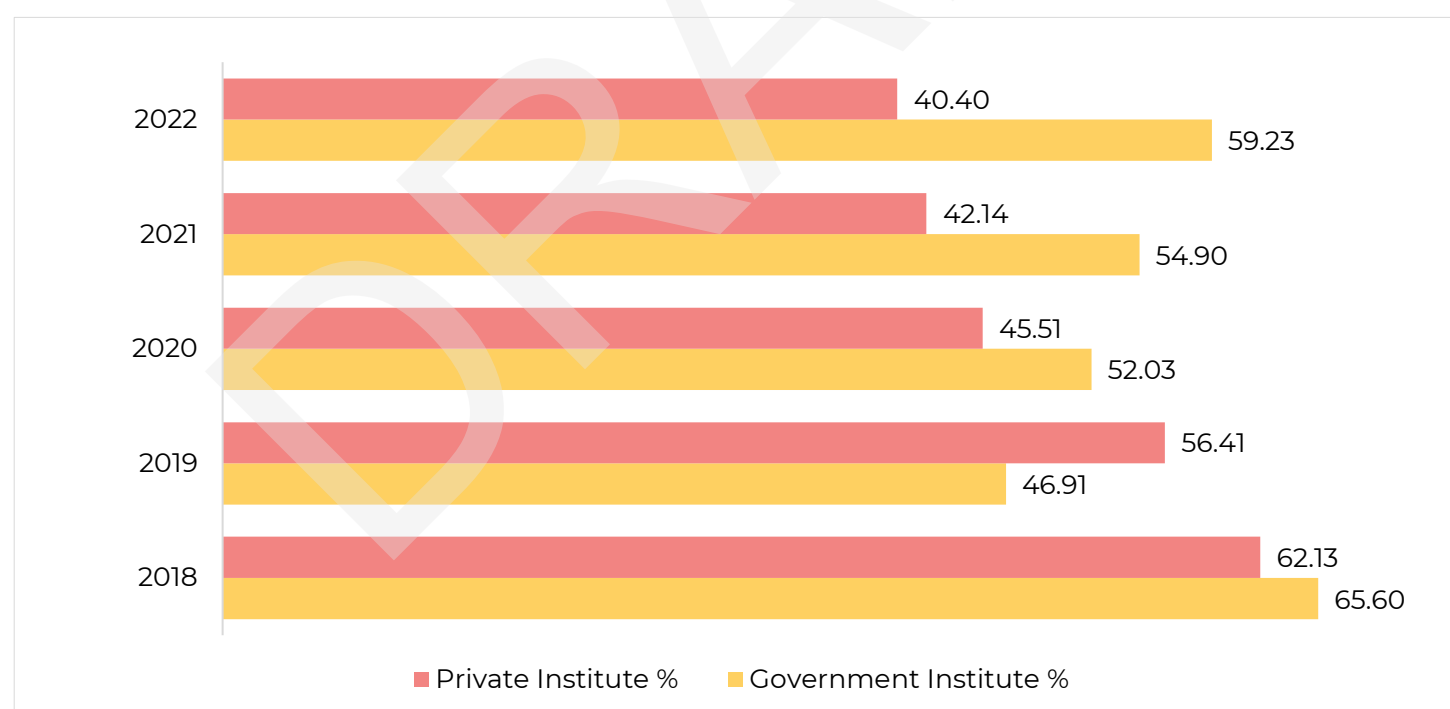


Figure 11: Seat utilisation across Government and private ITI institutes in India

- Seat utilisation based on enrolled candidates in ITIs is low with Government ITIs having better seat utilisation compared to private ones.
- In year 2022, **~40% seats** in Government ITIs are unutilised while private ITIs have only **~40% of seats** utilised against their enrolment capacity. ⁸

⁷ Directorate General of Training (DGT) - NCVT MIS Portal – Gender Distribution. Retrieved 14 July 2023

⁸ Directorate General of Training (DGT) - NCVT MIS Portal – Seat Utilization. Retrieved 14 July 2023

Grading of ITIs

DGT under the aegis of MSDE has taken an initiative to grade all the ITIs across the nation on key identified technical and institutional parameters. The key objective of conducting grading exercise for ITIs is to identify the good performing institutes, reward them and provide an opportunity to the institutes lagging in performance for them to improve upon.

As of now, grading in ITIs has been conducted in two phases. Phase 1 exercise was initiated in November 2017 and **4,811 ITIs** participated in this phase voluntarily. In Phase 2, **12,352 ITIs** have been graded on identified **27 parameters** categorised in 5 buckets each having 20% weightage.⁹

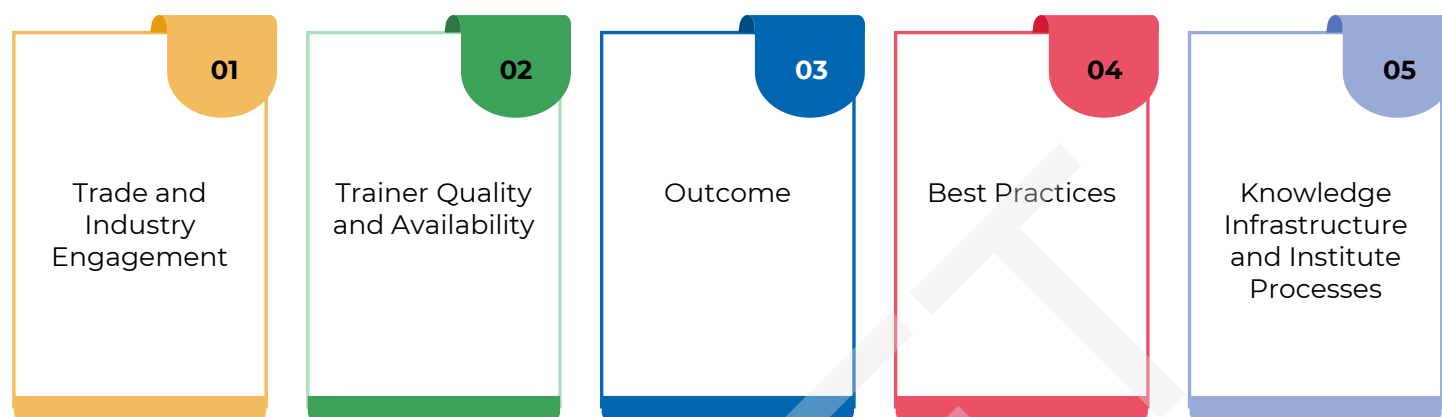


Figure 12: Broad five buckets for grading parameters of ITIs

Highlights of Phase 1 Grading of ITIs¹⁰

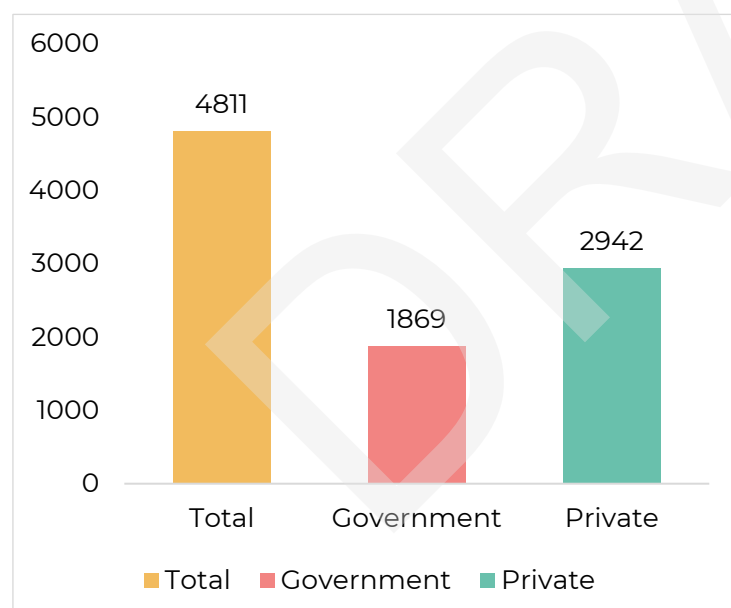


Figure 13: Participation from Government and private ITI institutes in phase 1 grading

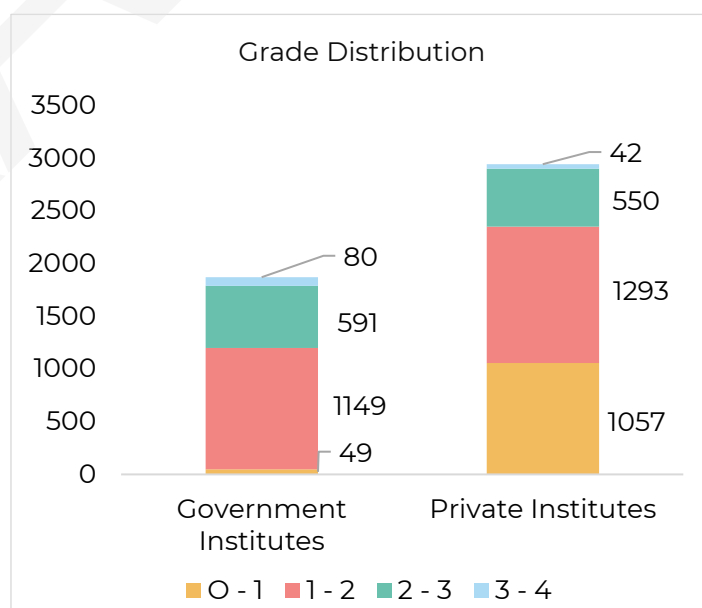


Figure 14: Grade distribution of Government and private ITI institutes in phase 1 grading

- Of the **4,811 institutes that participated in phase 1, 2442 (50.76%) institutes** have been graded on score range of (1 – 2), which is considered as low.
- Only **122 (2.54%) institutes** have been graded on score range of (3-4), considered as highest.
- **Tamil Nadu** has maximum **40 institutes** in the score range of (3-4) followed by **Gujarat** with **13 institutes** in this score range.

⁹ Directorate General of Training (DGT) – Grading of ITI. Retrieved 15 July 2023

¹⁰ Directorate General of Training (DGT) – Phase 1 Grading. Retrieved 15 July 2023

Highlights of phase 2 grading of ITIs¹¹

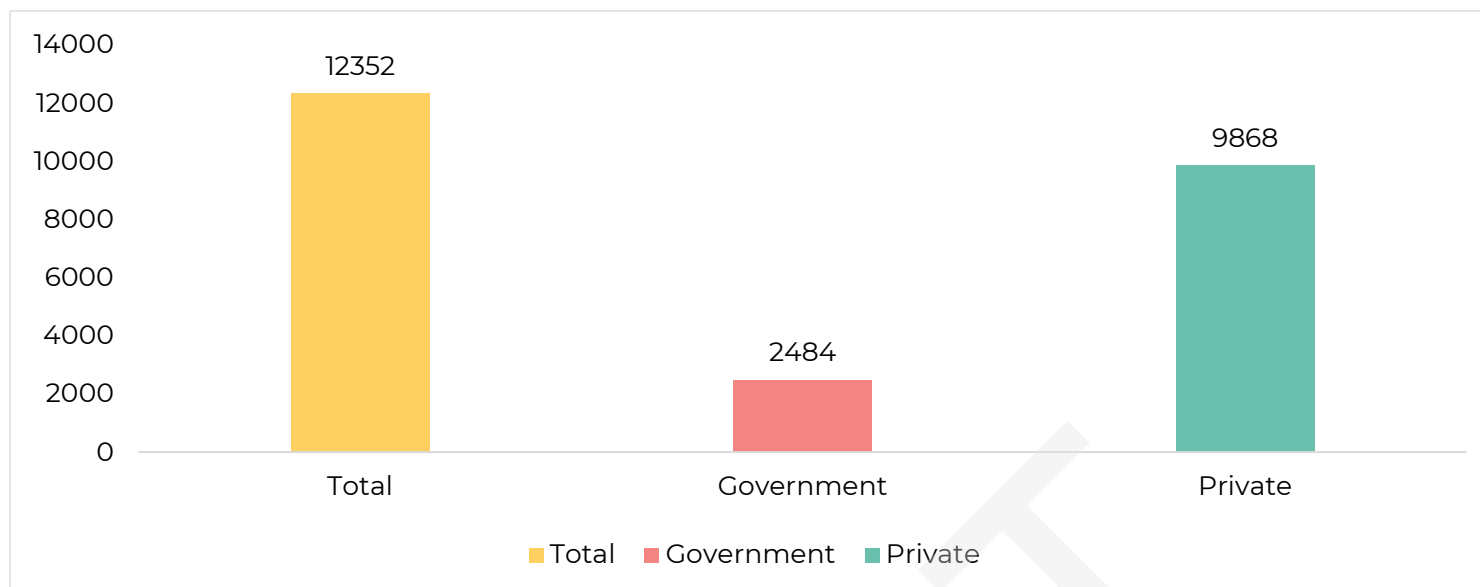


Figure 15: Participation from Government and private ITI institutes in phase 2 grading

- Of the **12,352 institutes** that participated, **5,238 (42.41%) institutes** have been graded on score range of (1 – 2), considered as low.
- **Only 237 ITIs (~2%)** have been graded on score range of (3-5), considered as high.
- **Andhra Pradesh** has maximum **43 institutes** in the score range of (3-5) followed by **West Bengal** with **34 institutes** in this score range.

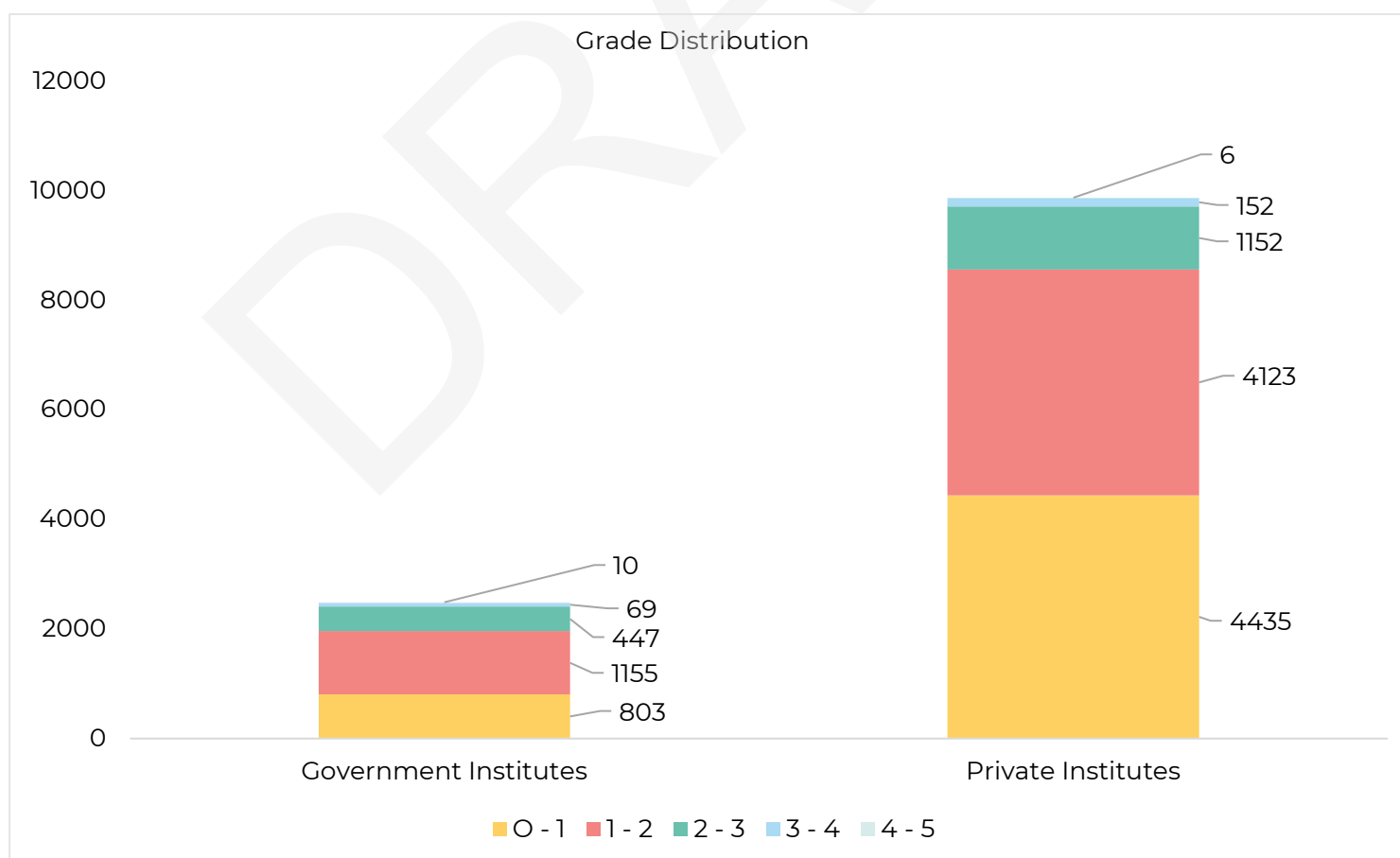


Figure 16: Grade distribution of Government and private ITI institutes in phase 2 grading

¹¹ Directorate General of Training (DGT) – Phase 2 Grading. Retrieved 15 July 2023

Accreditation of ITIs

Prior to 2012, accreditation of ITIs was done through Directorate General of Employment & Training (DGE&T) as per norms laid by National Council of Vocational Training (NCVT). Post 2012, Quality Council of India (QCI) was entrusted to manage the accreditation process. However, since 2017 the process has been directly managed by DGT.

Polytechnic Institutes – State wise distribution

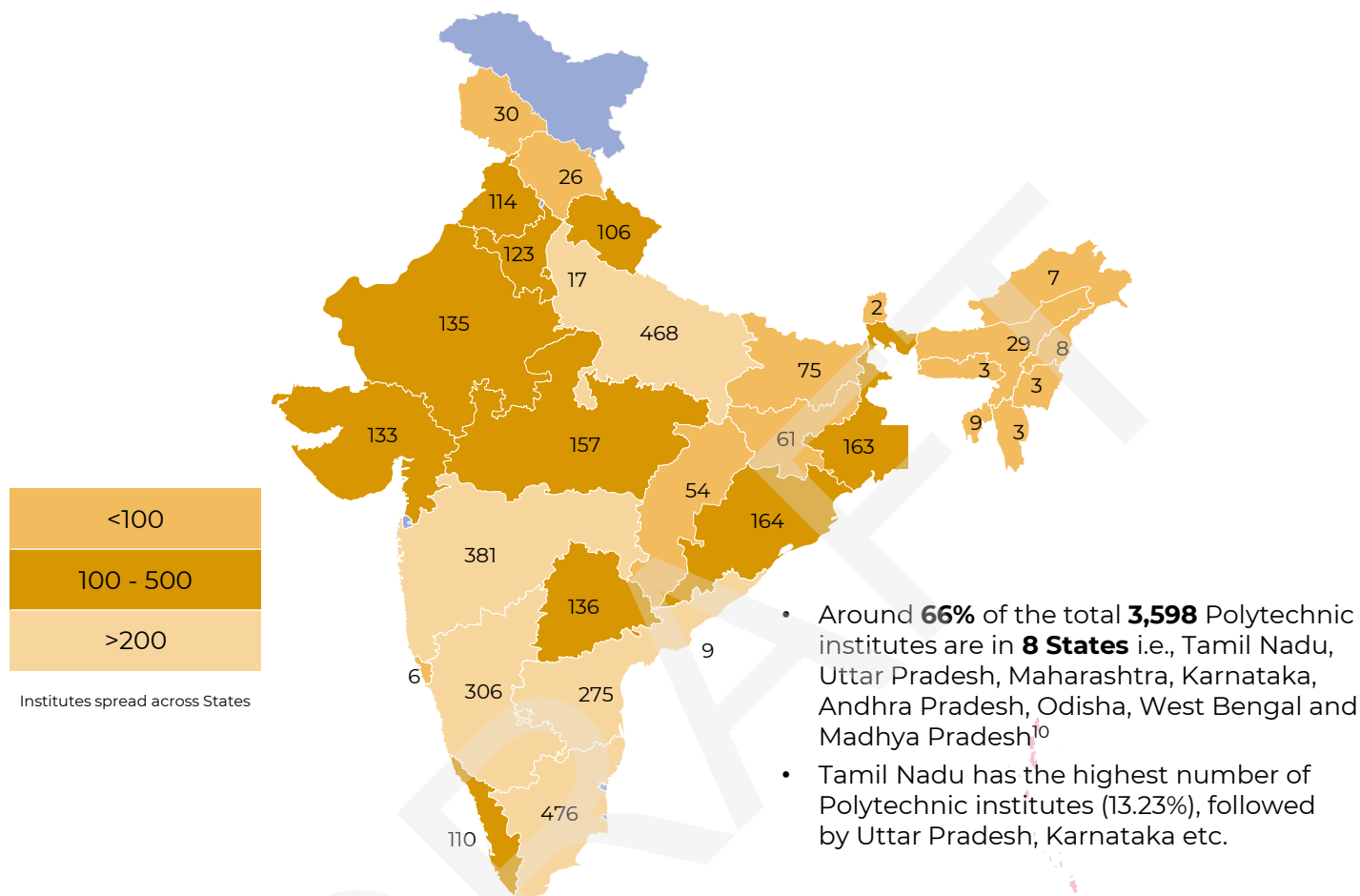


Figure 17: Polytechnic institutes spread in India across different States¹²

Affiliated Polytechnic Institutes across last five Academic years¹³

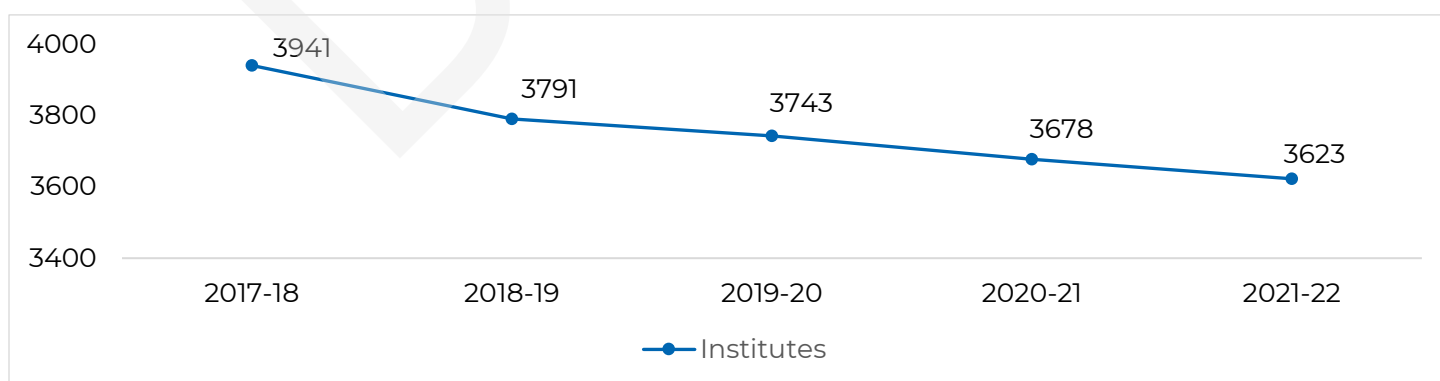


Figure 18: Polytechnic institutes across last 5 academic years in India

Over a period of last five academic years, several Polytechnic institutes have been disaffiliated across the country for not meeting the affiliation norms of AICTE. This has ensured that Government is focused on affiliating institutes who consistently meet the affiliation norms and have enhanced infrastructure availability, thus providing quality technical education to students.

¹² All India Council of Technical Education (AICTE) – Polytechnic institutes. Retrieved 25 July 2023

¹³ All India Council of Technical Education (AICTE) – Polytechnic institutes Affiliation. Retrieved 25 July 2023

Enrolment Distribution – Gender¹⁴

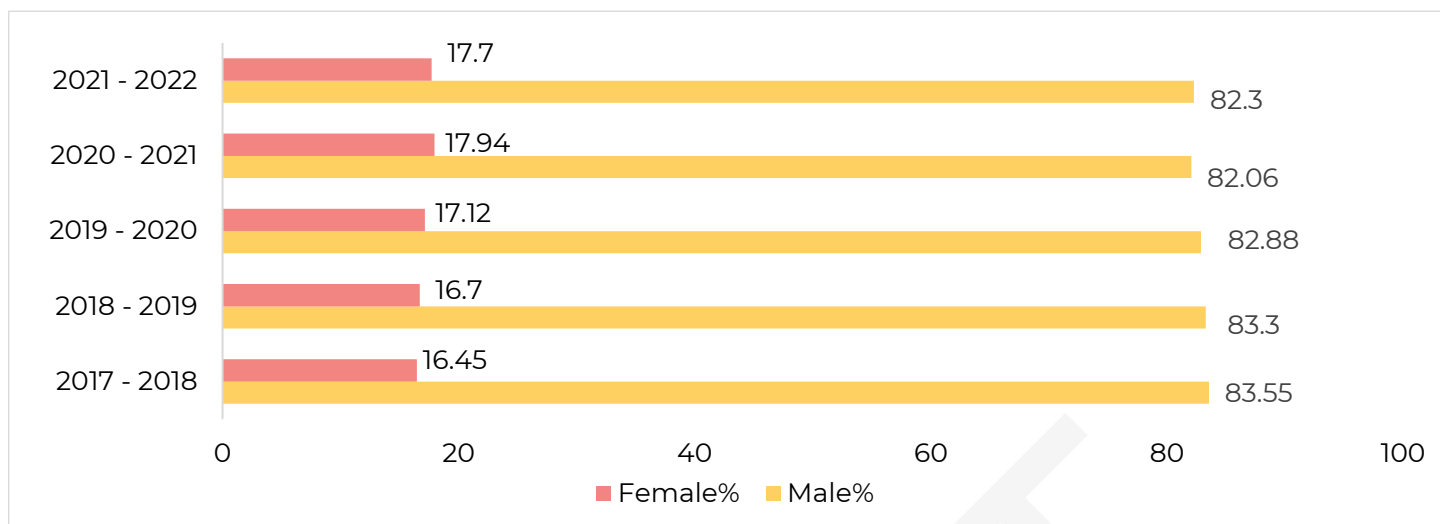


Figure 19: Enrolment distribution of Male and Female across Polytechnic institutes in India

- Enrolment of Male and Female candidates across Polytechnics in India has largely remained uniform over a period of academic years 2017 – 2018 to 2021 – 2022.
- There is a marginal increment of ~ **8%** in overall enrollment of female candidates during this period.

Institutional Trend in Polytechnic Institutes¹⁵

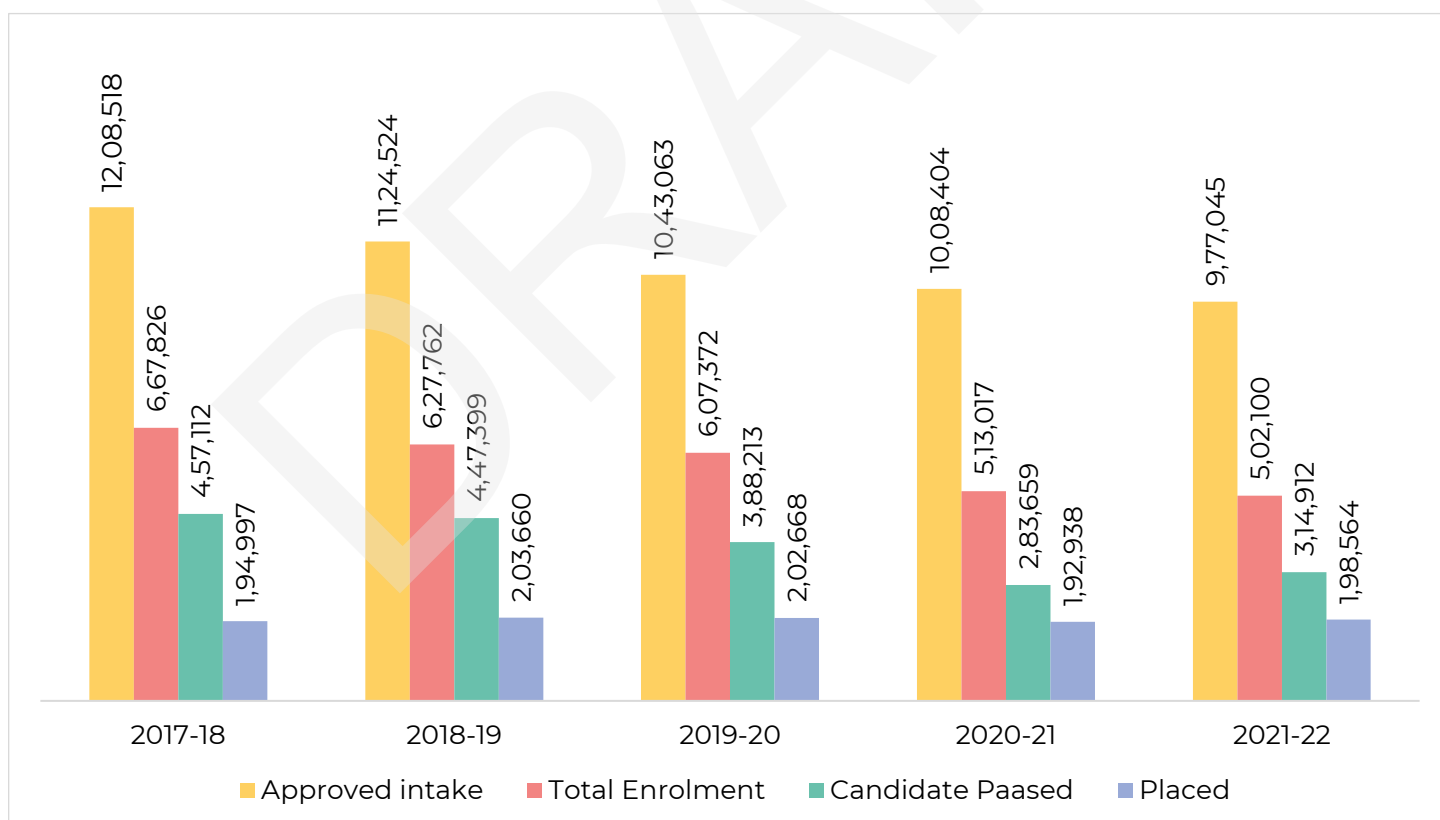


Figure 20: Candidate enrolment, certified and placement details in Polytechnic institutes across last 5 academic years in India

Over a period of last five academic years, the approved intake in Polytechnic institutes have seen decline due to disaffiliation of institutes. However, ratio for certified to placed candidates (in jobs) has significantly increased from **42.65%** in 2017-2018 to **63.05%** 2021-2022.

¹⁴ All India Council of Technical Education (AICTE) – Enrolment Distribution – Gender. Retrieved 12 July 2023

¹⁵ All India Council of Technical Education (AICTE) – Polytechnic Institutional trend. Retrieved 12 July 2023

Centre of Excellence: State-wise Distribution

With an aim to conduct high quality training in specific sectors with focus on emerging technologies and on developing an association between academia and industry for fostering skill training in country, MSDE has launched a scheme for recognition of Center of Excellence (CoE). As of now, 30 centers across 14 States in the country have been established/ recognised under the initiative.

CoEs recognised by MSDE¹⁶

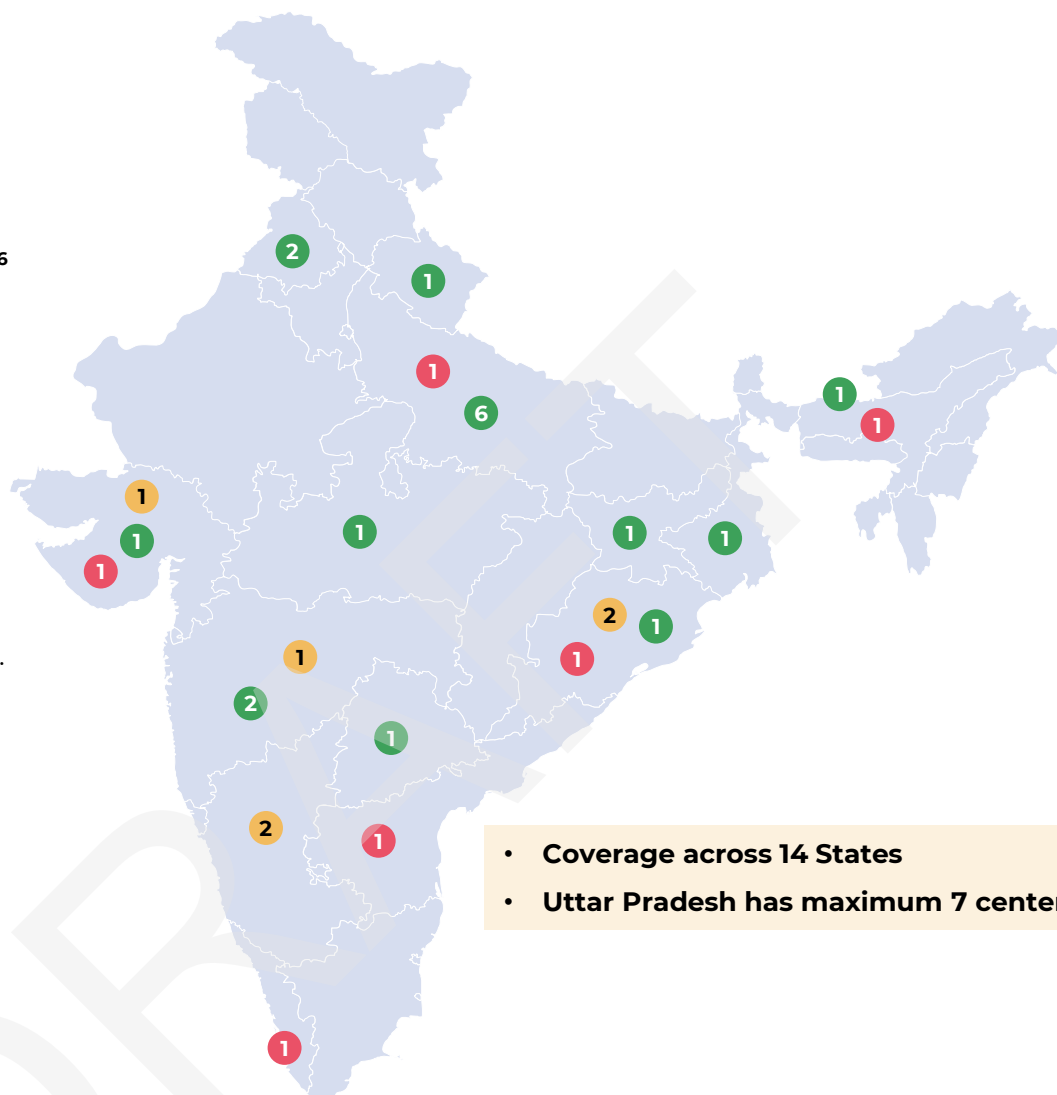
focus on key sectors such as Marine Engineering, Automotive, Electronics, IT-ITeS, Entrepreneurship etc.

Skill Development Institutes (SDIs)¹⁷

focus on sectors such as IT-ITeS, Automotive, Capital Goods, Logistics, Hydrocarbons, Electronics, Life Sciences, among others.

MSME Tool Rooms¹⁸

focus on General Engineering sectors.



- Coverage across 14 States
- Uttar Pradesh has maximum 7 centers

Figure 21: Centre of excellence institutes spread in India across different States

MSDE CoEs – 6

- Odisha (2) – SoVET, SIDAC
- Karnataka (2) – XIME, IWST
- Gujarat – EDII
- Maharashtra – Centre of Excellence, INS Shivaji

SDIs by PSU - 6

- Odisha (1) – SDI
- Gujarat (1) – SDI
- Kerala (1) – SDI
- Assam (1) – SDI
- Andhra Pradesh (1) – SDI
- Uttar Pradesh (1) – SDI

MSME Tool Rooms – 18

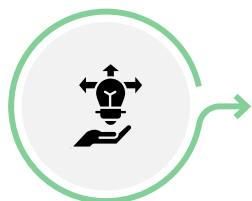
- Odisha (1) – CTTC
- Jharkhand (1) – IDTR
- West Bengal (1) – CTTC
- Assam (1) – TRTC
- Maharashtra (2) – IGTR, IDEMI
- Madhya Pradesh (1) – IGTR
- Gujarat (1) – IGTR
- Punjab (2) – CTR, CIHT
- Telangana (1) – CITD
- Uttarakhand (1) – ESTC
- Uttar Pradesh (6) – PPDC, CDGI, FFDC, CFTI

¹⁶ Ministry of Skill Development & Entrepreneurship Annual Report 2022 – 23

¹⁷ Ministry of Petroleum and Natural Gas - Skill Development. Retrieved 27 July 2023

¹⁸ Ministry of Micro, Small and Medium Enterprises - Operational Technology Center. Retrieved 27 July 2023

Opportunity and Challenges



Opportunities

- ITIs and Polytechnics could map their training programs with sectoral economic investments in the region or State where these institutes are located to enhance employment opportunity for candidates.
- ITIs and Polytechnics could strengthen their curriculum and teaching practices in alignment with the needs of new technology areas which would translate into sustained availability of job-ready workforce.
- Clear education and learning pathways for learners from ITIs and Polytechnics enabling progression to higher education courses, apprenticeships etc.
- Use of ITI and Polytechnic infrastructure for upskilling programs for industry workforce, job oriented short-term training programs, setting up production cum training centers and expansion of vocational education in schools (as per provisions of NEP).
- Innovative and sustained private sector engagement models in ITIs for enhanced quality and relevance of training programs leading to better placements. For instance, The Private Training Partner (PTP) model has been implemented by Government of West Bengal in their ITIs.



Challenges

- Capacity utilisation is one of the key challenges for both ITIs and Polytechnics, leading to unutilised infrastructure. In year 2022, **~40% seats** in Government ITIs are unutilised while private ITIs have only **~40% of seats** utilised against their enrolment capacity.¹⁹
- Industry relevance of training programs from institutions is low due to limited industry involvement in institute's functioning.
- Mismatch between industry expectation and competency level of trainees. During academic year 2021-22, **~40% candidates from polytechnic institutes** got placement.²⁰
- Limited focus on soft and employability skills of trainees.
- Inadequate availability of skilled trainers causing overburden on existing ones.
- Infrastructure quality and availability related challenges make the practical training less effective.
- Inadequate mechanism for review and updation of existing courses as per market demand.
- Performance of institute staff is not measured by placement of trainees or other output indicators around student performance leading to low or even no incentive to drive outcomes at institute level.
- Limited control at institute level on issues impacting the training outcome.

¹⁹ Directorate General of Training (DGT) - NCVT MIS Portal – Seat Utilization. Retrieved 14 July 2023

²⁰ All India Council of Technical Education (AICTE) – Polytechnic Institutional trend. Retrieved 12 July 2023

Cross-cutting Themes



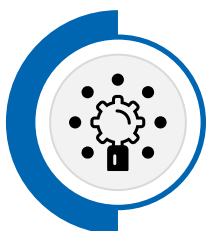
Demand aggregation and alignment with industry needs

- Industry partnership by few States like Karnataka, Assam, West Bengal, Uttar Pradesh for infrastructure upgradation of select ITIs and training of candidates based on the workforce demand.
- Diploma in IT, Networking and Cloud domain is being offered along with IBM in **40 ITIs and National Skill Training Institutes (NSTIs)** to cater to the need of industry demand covering **13 States**.



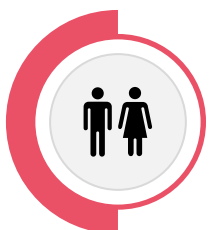
Future of work

- Center of excellence/ advanced centers focused on training of candidates emerging areas
- Alignment of curriculum at ITIs and Polytechnic in line with industry demand of future of work.



Digitisation of skills

- Recognising the need and importance of employability skills, DGT introduced “Employability Skills” module with different variants of learning hours in curriculum.
 - 80 Hrs. for all CTS trades of six months duration
 - 160 Hrs. in 1st year for all one- and two-year CTS trades
 - 80 hrs. web-based module in 2nd year for 2 years trade only
- DGT has launched ‘BharatSkills’ digital platform for accessing online content of different trades in ITIs along with option of blended learning in 15 course categories across ITI institutes.



Gender and social inclusion

- DGT has launched scheme for Skill Development in 48 districts across 10 States affected by Left Wing Extremism (LWE).
- Vocational training program for women is being offered through 19 National Skill Training Institutes for Women - NSTIs(W).
- State Governments like Goa, Gujarat, Haryana, Tamil Nadu among few others have launched schemes to empower marginalized community/ persons with disabilities. For example, Project Sankalp by Gujarat Skill Development Mission.

Government Interventions

Major Government Schemes for ITI Institutes

Since its inception, DGT has focused on and launched multiple schemes for revamping and improving the efficacy of training in ITIs. Significant efforts have also been taken for infrastructure upgradation of the institutes. A snapshot the same is mentioned below.

Scheme Name/ Parameters of Scheme	Launched By	Launch Year	Scheme Focus	Type of Beneficiaries	Minimum Eligibility Qualification	Training Duration
Craftsman Training Scheme	MSDE	1950	Employment linked vocational training	Any youth seeking vocational training	8 th Pass	6 Months – 2 Years
Advanced Vocational Training Scheme	Erstwhile DGE&T, Ministry of Labour and (now MSDE)	1977	Skill upgradation	Industrial workers	-	1 Week
Advanced Vocational Diploma	DGT, MSDE	2019	Technical Diploma	Youth seeking technical diploma	10 th /12 th pass with National trade certificate	2 Years
ESDI North-Eastern States	DGT, MSDE	-	Infrastructure upgradation Skill training	Youth from northeast region	8 th Pass	6 Months – 2 Years
Skill Development in districts affected by LWE	DGT, MSDE	2010 – 2011	Infrastructure upgradation Skill training	Youth from LWE affected districts	8 th Pass	6 Months – 2 Years
Upgradation of Govt ITIs through PPP	Erstwhile DGE&T, Ministry of Labour and (now MSDE)	2008	Infrastructure upgradation	NA	NA	NA
Skills Strengthening for Industrial Value Enhancement	MSDE	2017	Improving the relevance and efficiency of skills training through ITIs	Youths from ITIs	NA	NA

Upgradation of Govt ITIs into Model ITI	DGT, MSDE	2014	Infrastructure upgradation, Training of unorganized sector workers,	Youths from ITIs	8th Pass	6 Months – 2 Years
Grading of ITIs	DGT, MSDE	2017	Benchmark for comparison	NA	NA	NA

Table 1: Schemes for improving the efficacy of training and infrastructure in ITI institutes

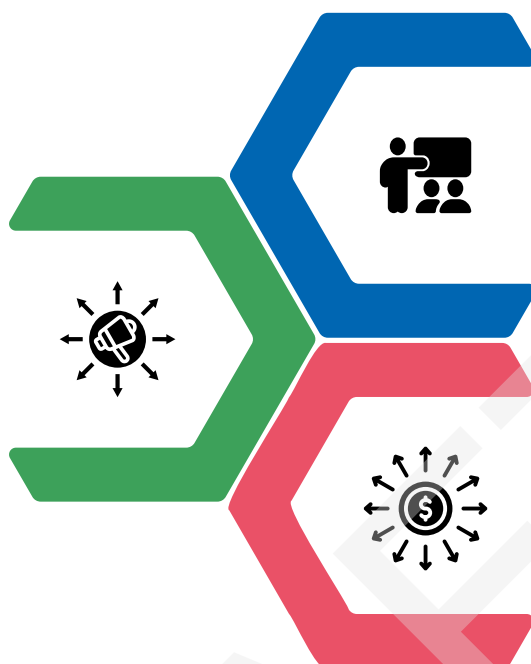
Major Government Schemes for Polytechnic Institutes

Scheme Name/ Parameters of Scheme	Launched by	Launch Year	Scheme Focus	Type of Beneficiaries	Minimum Eligibility Qualification	Training Duration
Community Development through Polytechnics (CDTP)	Erstwhile Ministry of Human Resource Development (now MSDE)	1978-79	Short term, employment-oriented skill development program	Any youth seeking vocational training	10th Pass	3 Months – 6 Months
Setting up of new polytechnics in un-served & underserved districts	Erstwhile Ministry of Human Resource Development (now MSDE)	NA	Infrastructure creation	Industrial workers	NA	NA
Central assistance for construction of women's hostels in selected polytechnics	Erstwhile Ministry of Human Resource Development (now MSDE)	NA	Enhance women enrolment	Women candidates	NA	NA
Central assistance for up-gradation of selected Polytechnics	Erstwhile Ministry of Human Resource Development (now MSDE)	NA	Infrastructure upgradation	Any youth seeking vocational training	NA	NA

Table 2: Schemes for improving the efficacy of training and infrastructure in Polytechnic institutes

Access to Skilling

- Expansion and continuous upgradation of institutes with industry participation. Leveraging learning from States like Karnataka, Assam, West Bengal, Uttar Pradesh.



- Integration of medium to long term training ecosystem with educational institutes – linkage with National Credit Framework (NCrF). Candidates from ITIs and Polytechnic institutes will get recognition from higher educational institutes.
- Social inclusion – skilling for disadvantage groups and promotion of traditional skills training based on presence of industry cluster by extending existing schemes of DGT/ AICTE for impact at scale.
- Promoting continuous learning and making content available in vernacular language.

- Standardisation across Skill Ecosystem – eliminate difference in training standards, assessment and certification pattern for all courses being offered in ITIs and Polytechnic institutes.



- Scaling up the industry collaboration increased absorption of trained candidates.



Improving State capacity and governance of institutions

- Inviting industry partnership for expansion and upgradation of infrastructure. Inference can be taken from few pilot projects being implemented in States of Karnataka, Uttar Pradesh, Assam, West Bengal among few others. Such as **UDYOGA**, an innovative program launched by Government of Karnataka, to transform 150 Government ITIs into technology hubs.²¹
- Improve governance of institutions through data/evidence-based decision making by having centralised dashboard covering critical success factors. States like Madhya Pradesh, Maharashtra and Gujarat have developed integrated systems for data collection, reporting and monitoring.
- Inclusive approach and customised training for Divyangs, migrants, women and tribals. For instance, to encourage the participation of differently abled candidates in vocational educational institutes, the Government of West Bengal has started a dedicated ITI in Kolkata for Divyangs.²²
- Devising outcome driven performance management system for all institutional staffs.
- Enhanced autonomy to ITIs (academic and non-academic) for focused and sustainable growth.
- Course content to be translated in regional languages wherever applicable for course adoption at scale. Bharat Skills under the ambit of DGT is one such initiative.



Capacity building of trainers and improving teaching & learning

- Opportunity towards upskilling and reskilling for existing trainers to improve their technical and pedagogical skills on emerging industry trends. Several States have taken an initiative to tie up with premier educational institutes like IIM, IIT and others for capacity building of the instructors and staff of the TVET institutes.
- Skilling in new-age technology to manage the industry demand. Courses on IoT, Drone Technology etc. have already been introduced by DGT and more such advanced areas should be picked up for curriculum development and training.
- Augmented reality/ Virtual reality-based skill training using simulators to impart scenario based practical training leading to better learning outcomes.
- Leverage the established centers of excellence and its capacity to map the industry demand.



Increasing Industry Engagement

- Strengthening of Institute-Industry linkage for On-the-Job training and apprenticeship. Enhanced role of industry SMEs for delivering knowledge session/ guest lecture on identified topics.
- Exploring innovative models for industry engagement like training cum production centers or joint research and consultancy assignments should be explored for strengthening industry-academia linkages. Institutes like CTTC Bhubaneswar have implemented such models successfully.

²¹ UDYOGA. Retrieved 10 August 2023

²² Government Industrial Training Institute For Physically Challenged Boys' And Girls'. Retrieved 10 August 2023

Recommended Readings

- World Bank, (2023), Building Better Formal TVET Systems - Principles and Practice in Low- and Middle-income Countries
<https://documents1.worldbank.org/curated/en/099071123130516870/pdf/P175566037a5e20650a657068b5152205bf.pdf>
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